AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 12 · STUDENT LESSON

Logs traces and privacy

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Link the events. Minimise the record. Preserve only permitted evidence.

Construct useful diagnostic metadata while excluding private content and keeping trace linkability and control limits explicit.

Keep one habit: Logging policy → field-decision table → linked-span picture → value/control probes → mock builder → independent field selection.

MAIN LESSON · SECTION 1

Connect the diagnostic events without copying private content

> Useful tracing needs a declared purpose, fields, access and retention.

Learning goals

I can explain the **mechanism** and terminology of logs traces and privacy.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve preserved evaluation and diagnostic evidence

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Failure budgets and service objectives. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**
>
> _____
>
> _____
>
> _____
>
> _____

End goal

Explain logs traces and privacy using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of logs traces and privacy

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Choose fields for a purpose rather than dumping the request

> Metadata can help diagnosis while still needing access and retention controls.

Begin with a minimal denied-request record

A school service can record that request R7 was denied at the permission stage without copying the student’s passcode or question. The trace ID links related events; the stage and error category explain where the process stopped. Build the diagnostic record from the fields the policy actually permits.

Saving the complete request “just in case” changes the privacy consequences and is not required to preserve an honest failure report.

Define logs, trace linkage and minimisation

Logs support diagnosis, but excessive recording can expose private content. **Trace ID**s link stages without storing complete sensitive **input**s.

Complete allowed-field and handling policy

Main classroom logging policy	Purpose or constraint
trace_id	Connect events for one request; application-issued identifier, still potentially linkable
stage	Controlled category: request, authorize, lookup or validate
status	Controlled category: ok, denied or review
duration_ms	Supplied nonnegative integer duration; declare the span being described
error_category	Controlled non-sensitive category, not an arbitrary copied exception message
Excluded fields	Passcodes, raw student question/essay, teacher-answer text and names
Pictured retrieval extension	Its explicitly approved source_id metadata, such as C8, belongs only on lookup events
Access and retention	Declared classroom reviewer and deletion after the exercise; implementation not supplied
Scope	Fictional placeholders and mock records only; no actual private data or log store

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Repair the record by constructing the intended metadata

> Do not print the raw record while trying to demonstrate its removal.

Original main allowed fields and retention requirement

For this policy log **trace ID**, stage, status, duration and **error category**; do not log passcodes, student text or teacher answers. Restrict access and declare retention.

Complete field-by-field R7 decision

Main R7 fixture field	Decision under the declared policy	Reason
trace_id R7	Keep controlled identifier	Links stages without the full request
stage authorize; status denied	Keep controlled categories	Explains the stopping point
duration 3 ms, error permission_denied	Keep supplied mock metadata	Duration/category have declared types and purpose
Complete passcode value	Exclude field and value	Not needed for this diagnostic record
Student text or teacher answer	Exclude field and value	Outside the stated logging policy

Main R7 fixture field	Decision under the declared policy	Reason
Later comparison artifact	Keep sanitized diagnostic record	Honest history does not require retaining private raw content

State the policy before processing

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Logs traces and privacy — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A failure record includes trace R7, status=denied and the complete student passcode.	For this policy log trace ID, stage, status, duration and error category; do not log passcodes, student text or teacher answers. Restrict access and declare retention.	Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive error category.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change one logged field and predict the policy failure

> Keep the same diagnostic purpose and allowed-field rules.

Current task and response

- A failure record includes trace R7, status=denied and the complete student passcode. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the full-question picture change

- Change: A developer logs the full learner question.

▼ **Check after predicting**

Prediction: That additional field violates this logging policy.

Reason: Debugging convenience does not authorise storing more personal information.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Follow trace and source identifiers across the pictured stages

> Keep span duration separate from averages and from overlapping component times.

Main unsafe record and policy-compliant repair

A failure record includes trace R7, status=denied and the complete student passcode.

Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

Preserve only the corrected diagnostic record

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain a policy-compliant diagnostic record without passcodes, student text or teacher answers, so a later revision can be compared honestly.

Separate pictured trace, source and complete timing scope

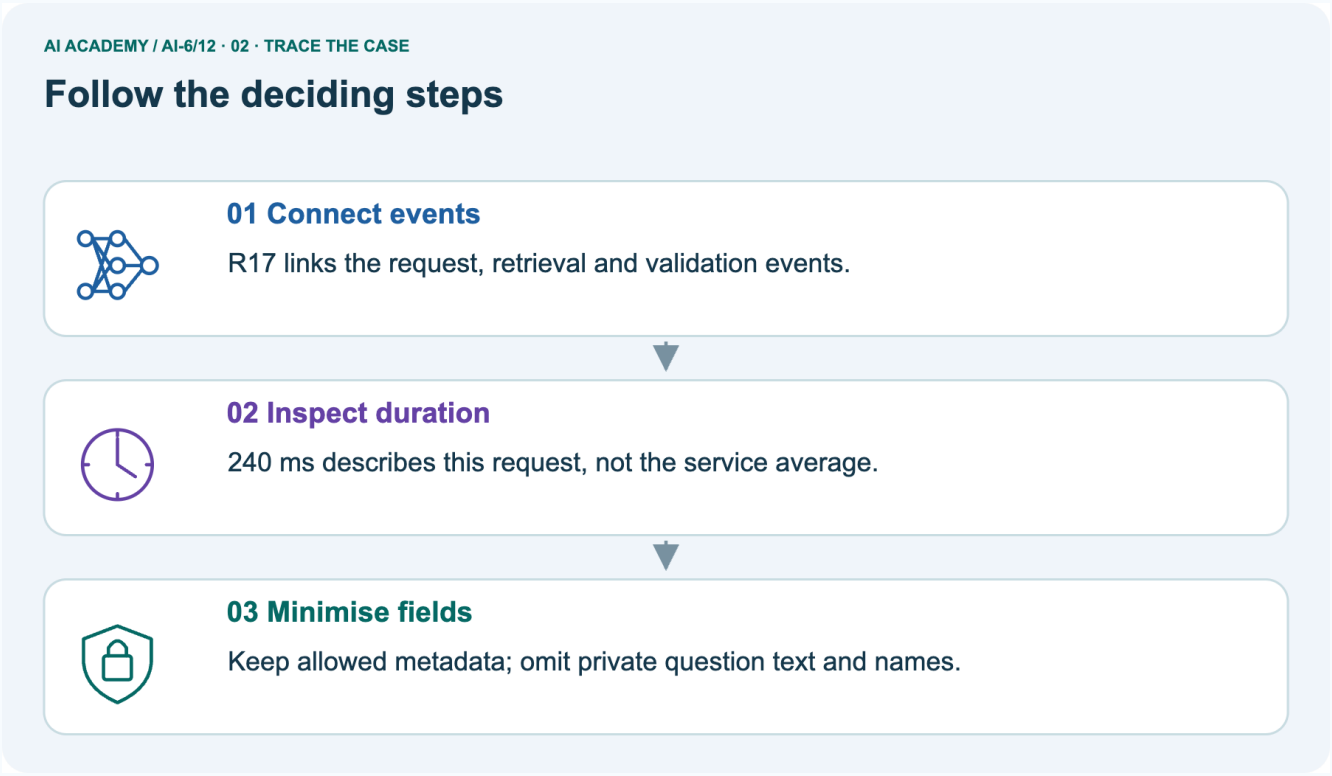
Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17.

R17 is the request **trace ID**; C8 is the retrieved source ID. Allowed fields also include validation pass and duration 240 ms. Raw question text is excluded.

The main denied request uses trace R7. The picture is a separate request R17 with retrieved source C8 and supplied request duration 240 ms. R17 connects related events; C8 names the source under this pictured metadata policy; 240 ms describes one request span, not the service average. The local extension uses mock lookup 80 ms and validation 20 ms spans linked to R17.

They can be nested inside the request span, so do not add them to 240 as independent extra work. No timing is actually measured by the example.

Trace related events and permitted fields



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Connect events	R17 links the request, retrieval and validation events.
2. Inspect duration	240 ms describes this request, not the service average.
3. Minimise fields	Keep allowed metadata; omit private question text and names.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Logs record events; traces connect related events. Useful debugging fields should remain within the declared privacy policy.

One trace links events without carrying the learner question

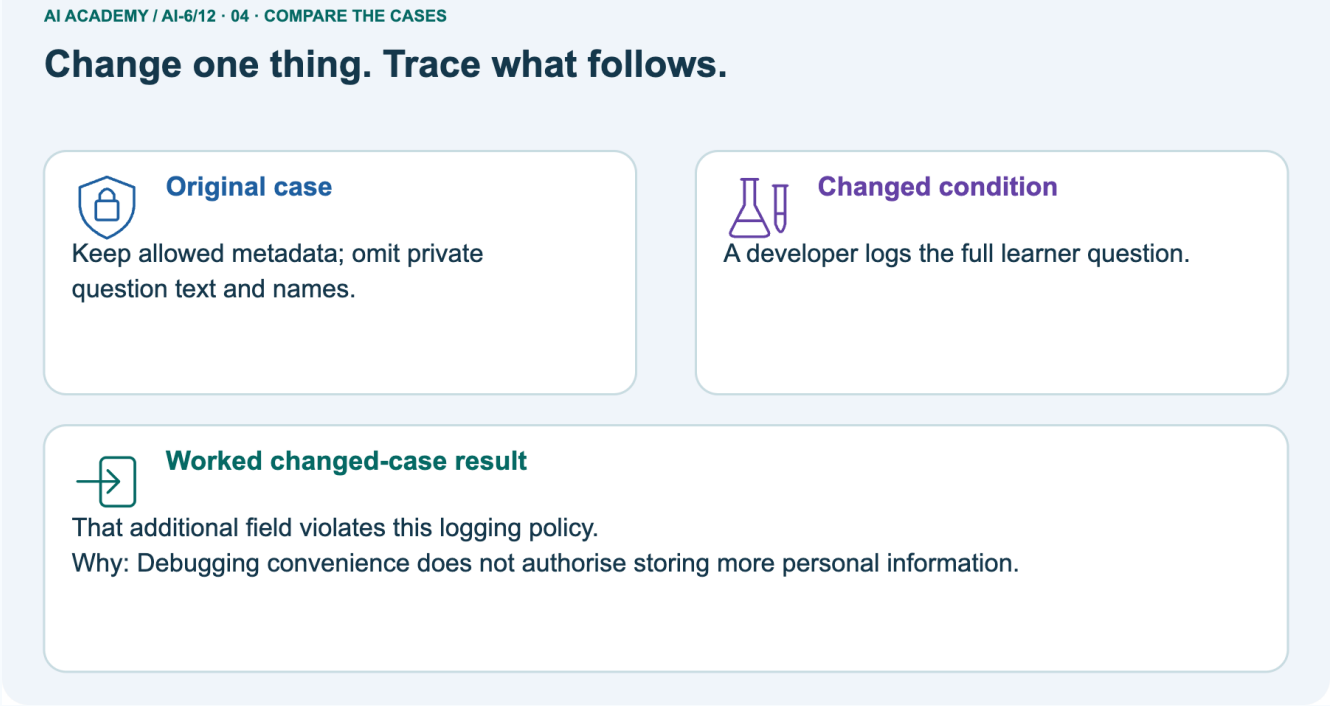
EVENT	PERMITTED METADATA	WHAT REMAINS EXCLUDED
Request span	R17, request, ok, supplied240 ms	Raw question and learner name
Lookup span	R17, lookup, ok, supplied80 ms, source C8 under declared extension	Retrieved teacher-answer text
Validation span	R17, validate, ok, supplied20 ms	Raw answer or arbitrary private error content

Supplied classroom spans, not real timing observations. A shared identifier is useful and linkable; it is not a guarantee of anonymity.

Predict adding the full learner question

- Change one thing: A developer logs the full learner question. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain omitted content and linkability limits

- **Trace id**entifiers can be linkable too; specify their access, retention and deletion rules.
- Read the labelled picture: Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?

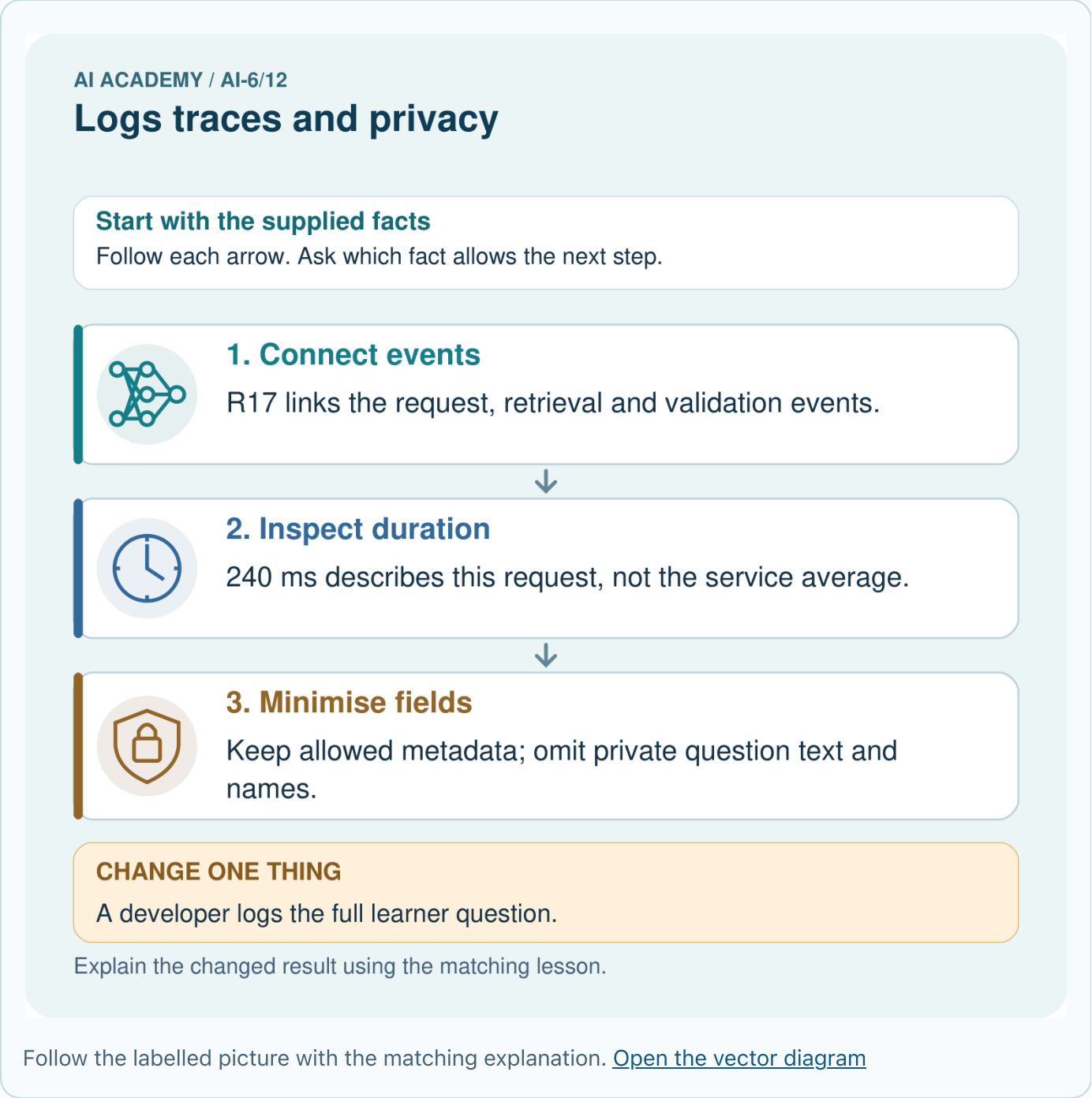
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Logs traces and privacy

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Minimise values as well as field names

> A trace ID can still link events to a person or private record through other data.

Core linkability and access qualification

A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete value, field and control-scope probes

Logging probe	Expected outcome or limit
Main R7 metadata only	Keep declared trace/stage/status/duration/error fields
Raw passcode, question or teacher text added	Violates policy; omit from emitted diagnostic record
Arbitrary private message in error_category	Reject; use declared non-sensitive categories
Boolean or negative duration	Reject under exact nonnegative-integer rule
C8 source ID attached to lookup event	Permitted only under the explicit pictured retrieval extension
Source ID attached to unrelated event	Reject in the local builder’s contract
A record has only allowed field names	Still validate values and purpose; identifiers can be linkable
Access/retention policy is printed	Declaration only; does not prove access enforcement or deletion occurred

Logging probe	Expected outcome or limit
One240 ms request fixture	Not the service mean, percentile or future latency guarantee

Guided explanation and repair

- A report omits this qualification: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise useful metadata and deliberate omissions

> Explain why a field is needed and why an excluded value is not.

Current explanation and guided scope check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example “Logs traces and privacy”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual linkability-limit check

- Use the worked case for Logs traces and privacy. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Trace-picture response

- Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?

> **My explanation using a pictured step and the given facts**

Reserve independent essay/guide field selection

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run the complete metadata-construction and emission audit

> Inspect only the intended emitted records, not an unsafe raw-data dump.

Current passcode-in-record investigation

A failure record includes trace R7, status=denied and the complete student passcode. Show the intermediate steps and give the conclusion with its scope.

Complete runnable mock builder and exclusion checks

```
"""Allowlisted metadata records from fictional fixtures; no real logs or secrets."""
import json
import re
POLICY={'access':'designated classroom reviewer',
        'retention':'delete mock records after this classroom exercise',
        'raw_text':'excluded'}

def event(trace_id,stage,status,duration_ms,error_category='none',source_id=None):
    if type(trace_id) is not str or not re.fullmatch(r'[A-Z][0-9]{1,8}',trace_id):
        raise ValueError('invalid trace ID format')
    if stage not in ('request','authorize','lookup','validate'):
        raise ValueError('invalid stage')
    if status not in ('ok','denied','review'):
        raise ValueError('invalid status')
    if type(duration_ms) is not int or duration_ms<0:
        raise ValueError('invalid duration')
    if error_category not in ('none','permission_denied','missing_source',
                              'invalid_request','dependency_failure','output_mismatch'):
        raise ValueError('invalid error category')
    record={'trace_id':trace_id,'stage':stage,'status':status,
            'duration_ms':duration_ms,'error_category':error_category}
    # The pictured retrieval policy explicitly permits this extra source-ID field.
    if source_id is not None:
        if stage!='lookup' or type(source_id) is not str or not re.fullmatch(r'C[0-9]
{1,8}',source_id):
            raise ValueError('invalid retrieval source ID')
        record['source_id']=source_id
    return record

raw_mock={'trace_id':'R7','stage':'authorize','status':'denied','duration_ms':3,
          'error_category':'permission_denied',
          'passcode':'fictional-passcode-placeholder',
```

```
          'student_text':'fictional private question',
          'teacher_answer':'fictional private guide text'}
# Build the intended record; never serialize or print the entire raw_mock.
safe=event(raw_mock['trace_id'],raw_mock['stage'],raw_mock['status'],
           raw_mock['duration_ms'],raw_mock['error_category'])
trace=[event('R17','request','ok',240),
       event('R17','lookup','ok',80,source_id='C8'),
       event('R17','validate','ok',20)]
serialized=json.dumps([safe,*trace])
assert set(safe)=={'trace_id','stage','status','duration_ms','error_category'}
assert all(raw_mock[key] not in serialized for key in
['passcode','student_text','teacher_answer'])
assert {r['trace_id'] for r in trace}=={'R17'}
assert trace[0]['duration_ms']==240 and trace[1]['source_id']=='C8'
print('Declared classroom policy, not implemented controls:',POLICY)
print('Main minimized record:',safe)
for row in trace:print('Separate pictured trace metadata:',row)
print('The 240 ms request duration is one supplied fixture value, not a service average.')
print('Request span can include lookup/validation spans; do not add nested durations as
independent work.')
print('Trace/source IDs remain linkable. Access control and deletion are declared, not
enforced by this builder.')
```

Expected metadata, validated fields and control limits

1. Predict the safe R7 record before running or tracing. The code uses fictional private placeholders but never prints or serializes the raw mock. It builds explicit metadata with controlled stages/statuses/error categories and validates duration/identifier shapes. It checks that the tested private placeholder values are absent from all emitted JSON.

   The pictured R17/C8 events are separate mock span records; no timer runs. The builder does not implement a real log store, authentication, retention deletion or universal privacy guarantee. Application-issued identifiers remain linkable; preserving failed-case evidence means a policy-compliant diagnostic artifact, not retaining raw private text.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	

Record	Your evidence
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: select permitted fields from a new mixed record

> Use the fixed policy and do not invent missing metadata.

Complete independent essay/guide/metadata task

Logs contain user essay, teacher guide, stage=lookup, duration=12 ms and trace T8. Select allowed fields.

First attempt and support record

> **My independent explanation, trace and limits**
>
> ---
>
> ---
>
> ---
>
> ---

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Logs contain user essay, teacher guide, stage=lookup, duration=12 ms and trace T8. Select allowed fields.

> **My independent procedure, result and limitation**
>
> ---
>
> ---
>
> ---
>
> ---

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the diagnostic schema with access and retention requirements

> Keep sanitized failure evidence and explicitly identify controls still unimplemented.

Project task and artifact

Add a labelled logs traces and privacy **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of logs traces and privacy.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a useful trace and no unnecessary private content

> A minimal record still requires controlled access, retention and honest measurement claims.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of logs traces and privacy

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Trace ID	An identifier connecting events within an operation
Minimisation	Keeping only data needed for the declared purpose
Error category	A non-sensitive label for a failure class

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Logs traces and privacy — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A failure record includes trace R7, status=denied and the complete student passcode.	For this policy log trace ID, stage, status, duration and error category; do not log passcodes, student text or teacher answers. Restrict access and declare retention.	Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive error category.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17.

By the end, I can	How I will show it
Explain the mechanism and terminology of logs traces and privacy	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17.
- Which labelled step matches this detail? Keep allowed metadata; omit private question text and names.
- 1. Inspect duration
- 2. Connect events
- 3. Minimise fields
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Minimise fields Keep allowed metadata; omit private question text and names. Logs record events; traces connect related events.

Useful debugging fields should remain within the declared privacy policy.

Changed-case question and optional hint

- Changed-case practice: A developer logs the full learner question. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. That additional field violates this logging policy.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: That additional field violates this logging policy. Debugging convenience does not authorise storing more personal information.

Repair and retain the record

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the **mechanism** and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Review Integration tests and fixtures (ai-6/9). Goal: Explain the **mechanism** and terminology of integration tests and fixtures. Short practice: Follow request N through the fixture and adapter.

Which source and item should appear in the reply, and what precisely fails when the adapter returns M?. Return to this week and explain what you changed.

Review Least privilege and tool scopes (ai-6/6). Goal: Explain the **mechanism** and terminology of least privilege and tool scopes. Short practice: Classify read S2, write S2 and read S5 as permitted or denied.

Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

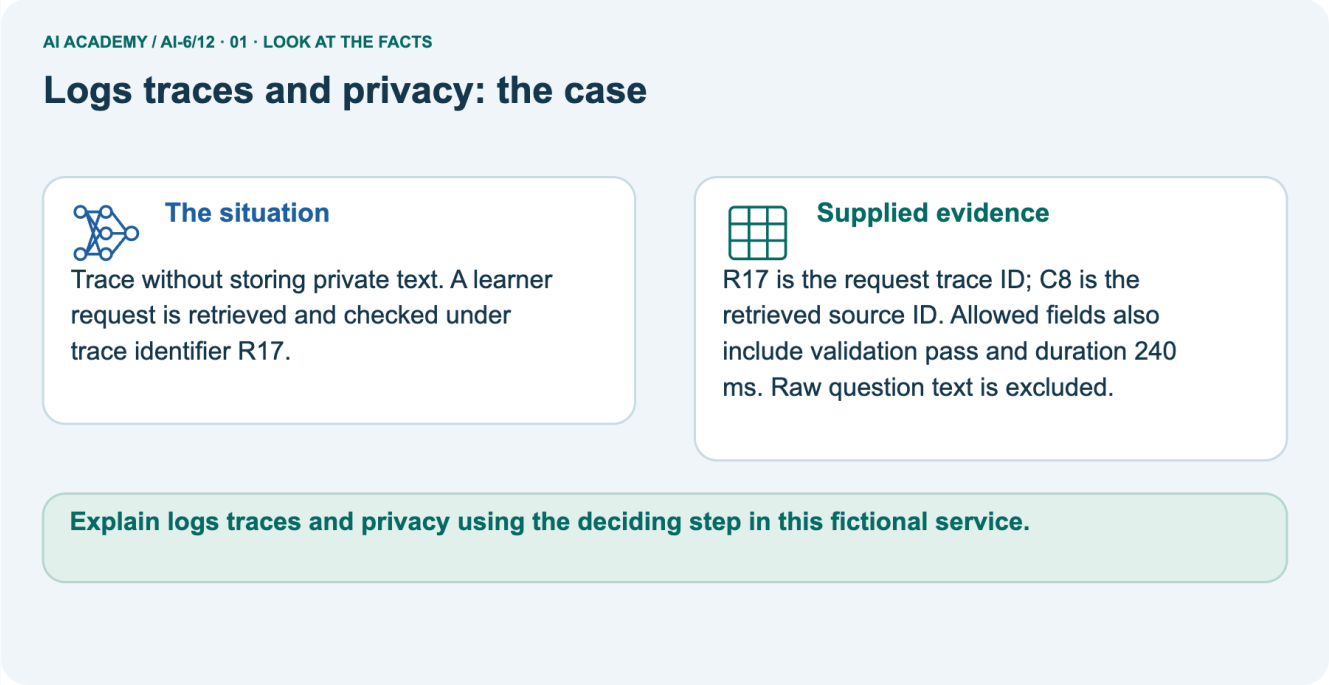
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Logs traces and privacy: the case



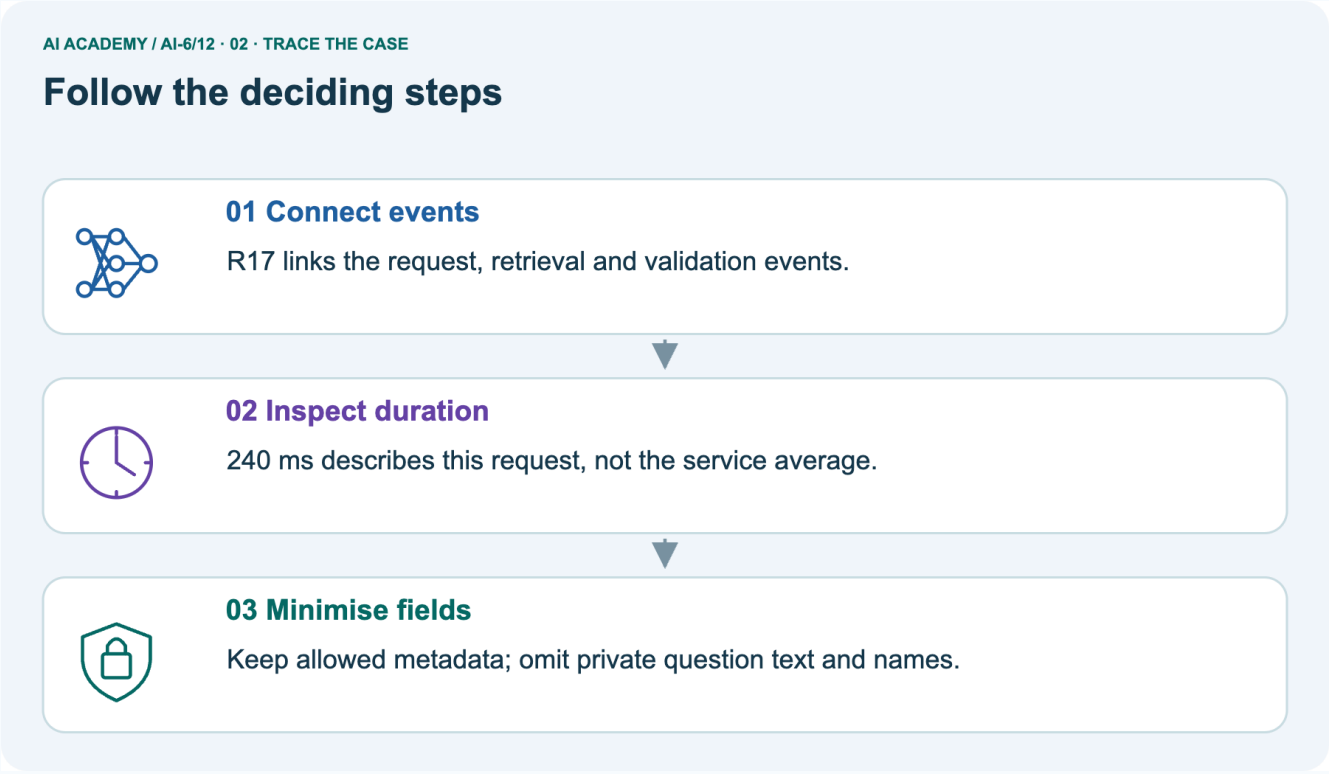
Follow the labelled picture with the matching explanation. Open the vector diagram

Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17. Supplied facts: R17 is the request **trace ID**; C8 is the retrieved source ID.

Allowed fields also include validation pass and duration 240 ms. Raw question text is excluded.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

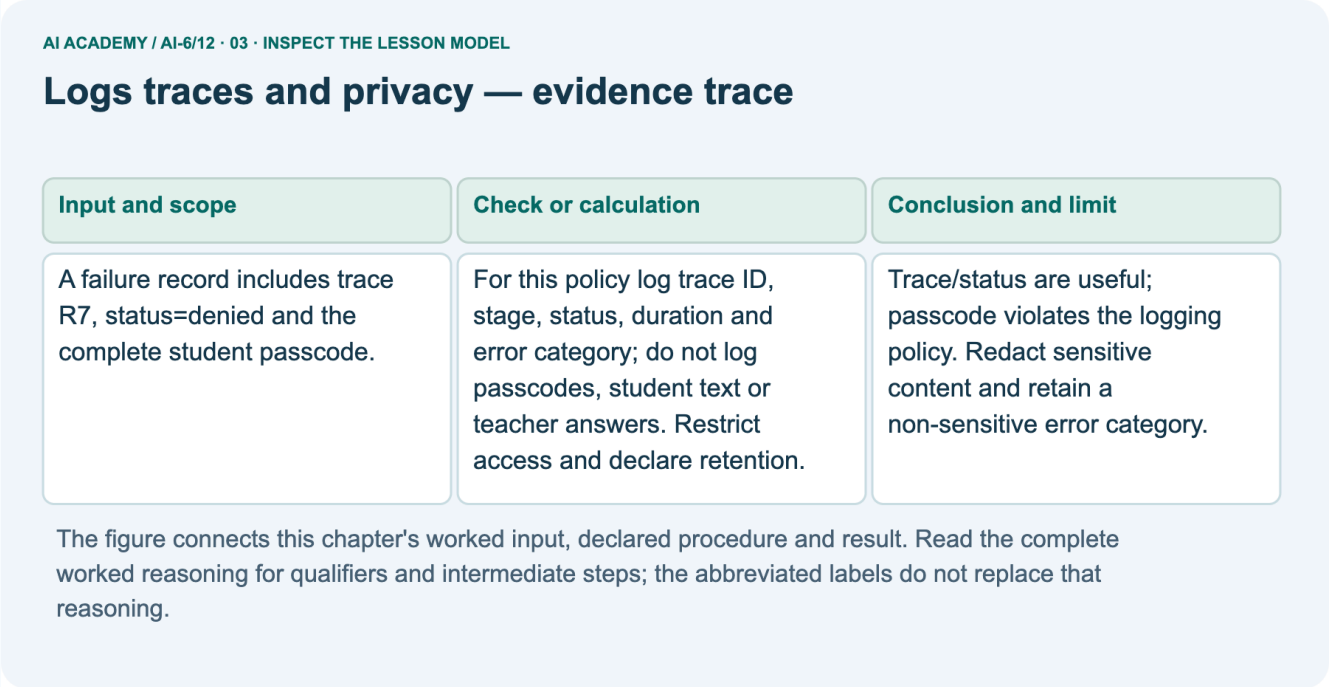


Follow the labelled picture with the matching explanation. Open the vector diagram

Connect events: R17 links the request, retrieval and validation events. → Inspect duration: 240 ms describes this request, not the service average. → Minimise fields: Keep allowed metadata; omit private question text and names.

Point to each numbered stage and explain which recorded **input** it uses.

03. Logs traces and privacy — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

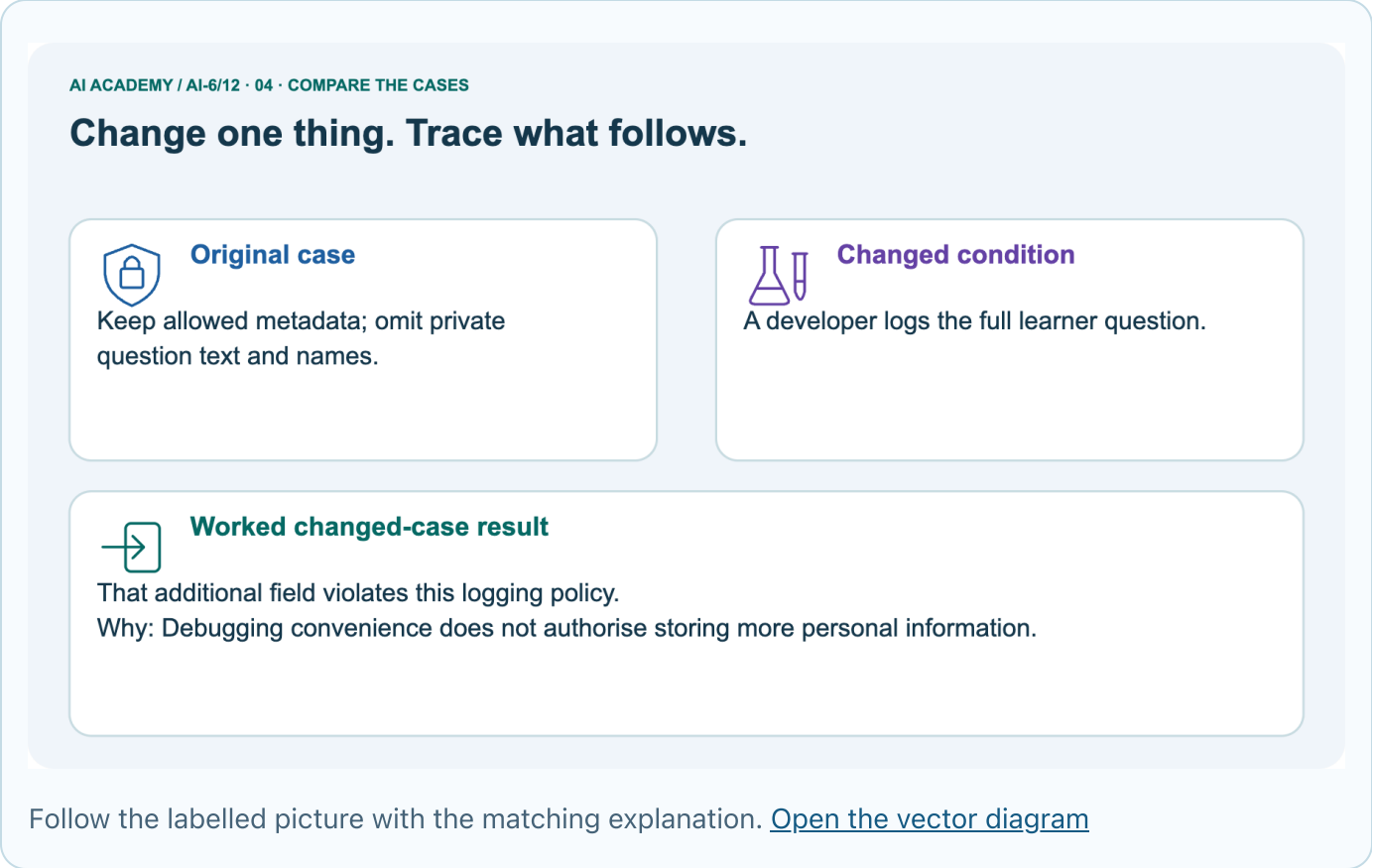
Logs traces and privacy — **evidence** trace.

A failure record includes trace R7, status=denied and the complete student passcode. | For this policy log **trace ID**, stage, status, duration and **error category**; do not log passcodes, student text or teacher answers.

Restrict access and declare retention. | Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



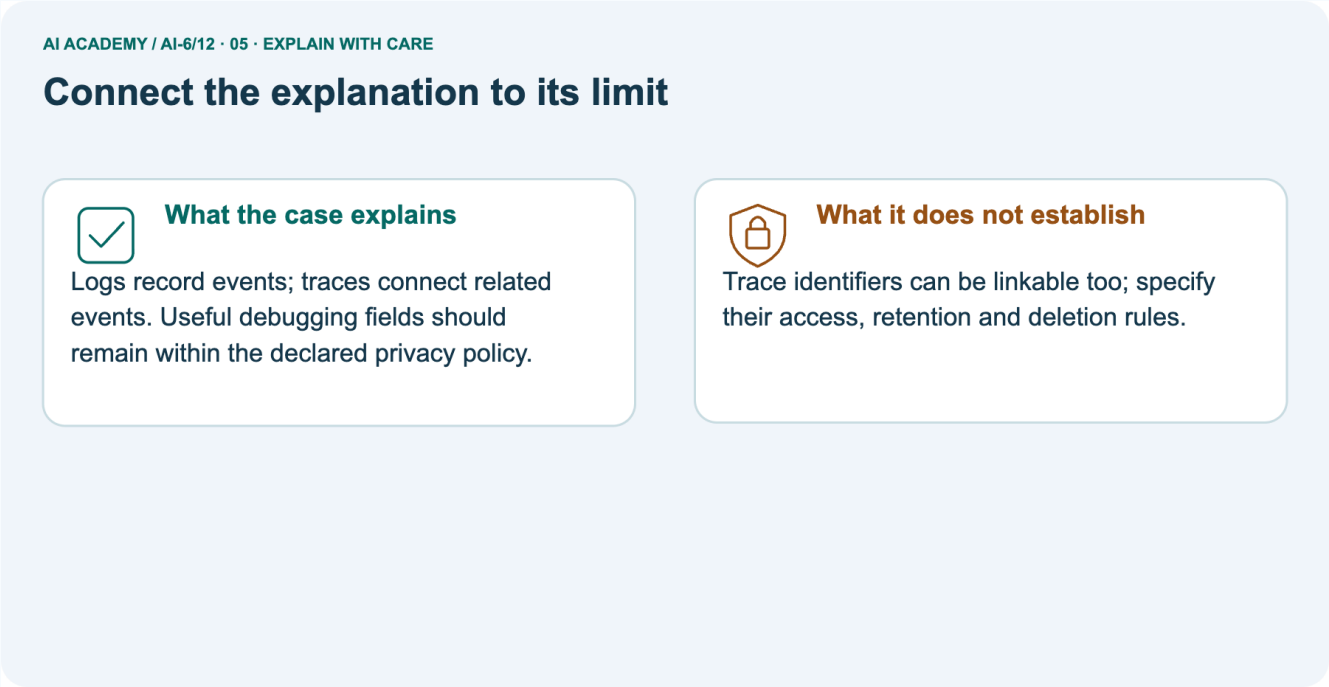
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep allowed metadata; omit private question text and names. Changed condition: A developer logs the full learner question. Expected result: That additional field violates this logging policy.

Reason: Debugging convenience does not authorise storing more personal information.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit

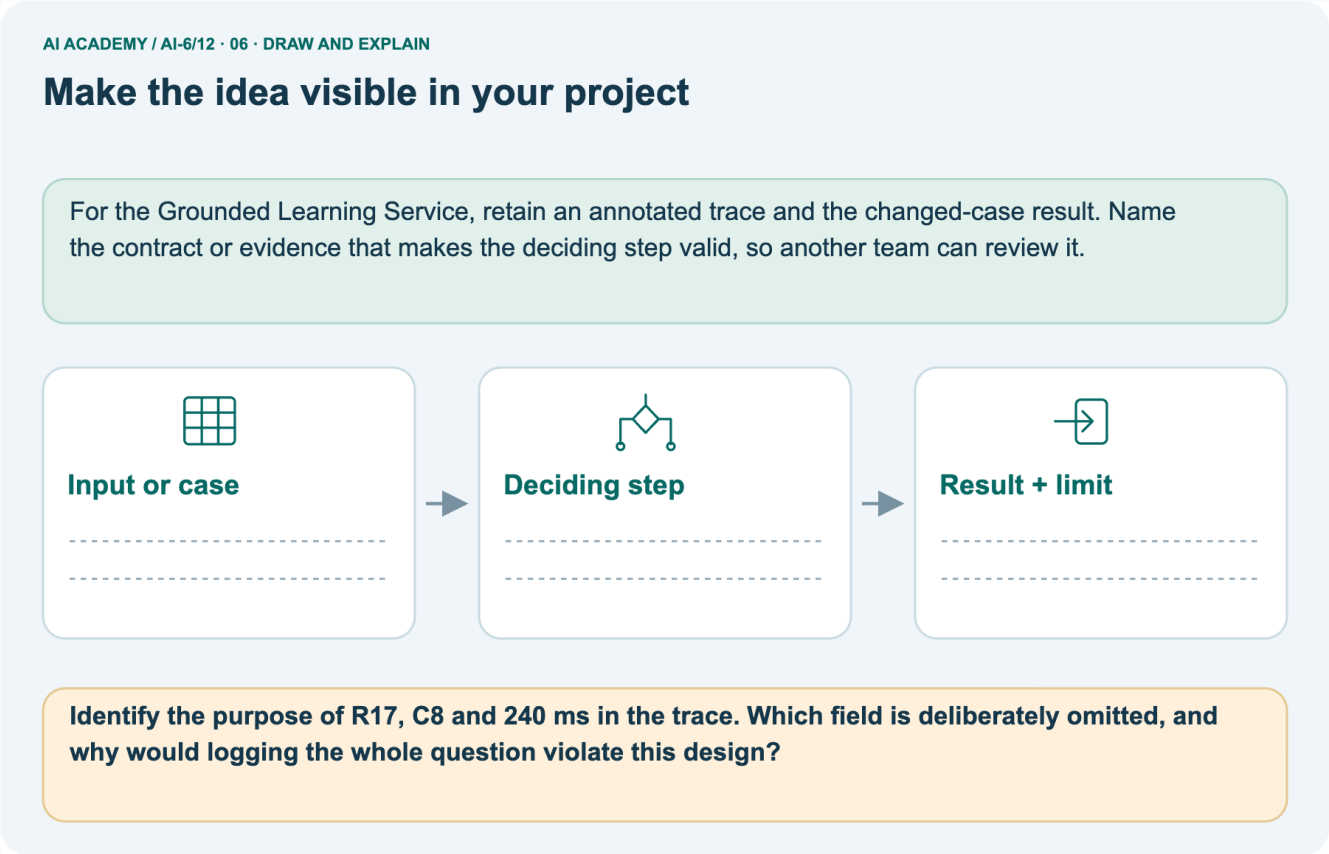


Follow the labelled picture with the matching explanation. Open the vector diagram

Logs record events; traces connect related events. Useful debugging fields should remain within the declared privacy policy. Limit: **Trace id**entifiers can be linkable too; specify their access, retention and deletion rules.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Logs traces and privacy: the case

02. Follow the deciding steps

03. Logs traces and privacy — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.